

Proactive connectivity handover in dynamic optical satellite networks

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The reduction in satellite launch costs has made the deployment of a large number of low Earth orbit (LEO) satellites possible. These satellites establish laser inter-satellite links (ISLs) to provide global coverage and high-speed data transmission. However, due to the mobility nature of satellite platforms, the wireless laser link is not as stable as the terrestrial fiber link. Laser ISLs are subject to predictable failures and random failures, both of which frequently cause connectivity handovers and transmission interruptions. To manage such challenges, this work tries to optimize the connectivity handover process with the aim of reducing the total interruption duration. We first define a dynamic satellite network model by introducing the time-to-disconnection risk for each ISL. Based on the network model, we design a proactive connectivity handover algorithm to determine the switching time and path for each handover operation, instead of waiting till the connectivity is interrupted. Simulation results demonstrate that the proposed proactive handover algorithm can effectively reduce the interruption rate by 41.37% and improve the maximum reliable transmission time by up to 31.89% compared to the baseline SP algorithm. © 2026 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

The recent reduction in space launch costs has significantly accelerated the development of LEO satellites [1], which typically operate at altitudes between 500 and 1500 km above the Earth's surface [2]. This trend is exemplified by initiatives such as Starlink, which has received approval from the U.S. Federal Communications Commission (FCC) to launch up to 12,000 satellites by 2027. Similarly, the OneWeb constellation has already deployed more than 648 satellites [3], and its planned second-generation system is expected to expand significantly, potentially reaching more than 6300 satellites in total.

To further enhance the performance of satellite constellations, satellites will no longer function as simple communication relays or point-to-point terminals. By establishing inter-satellite links (ISLs), they can form an interconnected network, enabling collaborative execution of complex tasks. For implementing ISLs, two primary technologies are available: radio frequency (RF) communication and laser-based communication. When considering antenna size and communication capacity requirements, laser technology stands out as a particularly promising solution. Its advantages, including ultra-high data rates and strong interference resistance, make it well suited for establishing ISLs. Networks that leverage such ISLs are commonly referred to as optical satellite networks [4].

Due to satellite operation in a public and uncontrollable space environment, optical satellite networks are susceptible to various failures. These failures can be broadly categorized into predictable and random failures [5]. Predictable failures mainly include intermittent ISL disconnection caused by solar eclipses [6], as well as ISL disconnection in polar regions due to pointing angles exceeding the payload's rotational limits [7,8], and so on. Random failures often involve ISL uncertainties resulting from space debris [9], among other potential issues. This work focuses on the predictable failures.

Such predictable failures will interrupt service transmission, necessitating effective strategies for prompt traffic recovery. In optical satellite networks, service transmission follows a circuit-switched model [10], which means that an end-to-end optical transmission path must be established before service transmission. Therefore, when ISLs are disconnected, a connectivity handover is required to establish a new end-to-end transmission path, resulting in transmission interruption. But frequent interruption not only increases recovery overhead, but also severely compromises service reliability in mission-critical applications, such as emergency response to disasters [11].

To mitigate the impact of handover on service continuity, several strategies have been proposed. In [12], the concept of grouped handover is introduced to address the slow routing

convergence caused by dynamic ISL states, thereby effectively reducing the route convergence time during handover. Reference [13] designs a novel handover architecture based on the predictability of satellite positions, which reduces partial signaling interactions and shortens the duration of interruption. Furthermore, in [14] the handover problem is formulated as a shortest-path search over a time-varying weighted directed graph, thereby optimizing the continuity of service transmission. Our previous work [15] proposed a post-failure handover strategy that computes a handover path based on ISL risk. However, post-failure handover struggles to avoid the direct impact of processes such as fault detection and rerouting on service interruption duration, exhibiting inherent limitations in optimizing single interruption duration. Besides, existing work concentrates on optimizing individual interruption time, with insufficient attention paid to the long-term total interruption duration from multiple handovers. Therefore, we further propose a proactive handover strategy, which aims to minimize the total interruption time of the service, thereby ensuring the continuity and stability of the optical satellite network.

In this work, we first analyze the factors that affect the total duration of interruption and clarify that it is jointly determined by the frequency of interruption and the duration of interruption for each handover. Therefore, the objective of this work is to reduce the total duration of the interruption by reducing the interruption time per interruption and minimizing the number of interruptions. To achieve the above objectives, we first construct a dynamic optical satellite network model from three dimensions: constellation topology, handover control procedure, and traffic distribution. Building on this model, we further propose a proactive handover algorithm.

This work is an extended version of [15], and the extra contributions include the following: (i) a proactive handover algorithm is proposed, which determines the switching time, thereby achieving proactive scheduling, and (ii) the path selection strategy of the proactive handover algorithm incorporates the asynchrony of service and introduces the concept of dynamic thresholds, based on the original scheme. Finally, we evaluate the proactive handover algorithm. First, the proactive handover mechanism not only effectively shortens the duration of interruption per handover, at the cost of increasing end-to-end latency, but the proposed algorithm can also reduce the overall service interruption ratio.

The rest of the paper is organized as follows: Section 2 discusses the model of optical satellite networks. Section 3 proposes the proactive connectivity handover strategy. Section 4 presents the simulation results and analysis. Finally, Section 5 concludes the paper and outlines future work directions.

2. DYNAMIC OPTICAL SATELLITE NETWORK MODEL

To address frequent connectivity handover problems in optical satellite networks, this section first introduces the constellation and topology design, handover control procedure, and traffic distribution model.

A. Satellite Constellation

Satellites typically operate in circular orbits [16,17], where the orbital planes and satellites are uniformly distributed, often employing the Walker constellation.

The constellation mainly has two configurations: Walker-Star and Walker-Delta. The Walker-Star constellation utilizes polar orbits, which offer the advantage of global coverage. Moreover, to reduce construction costs, the constellation is designed with a minimum number of satellites, ensuring maximum coverage [18,19]. The OneWeb constellation adopts this type of orbit. Our work primarily focuses on the Walker-Star constellation.

The constellation parameters can be succinctly represented as $\{alt, inc, N_{orb}, N_{sat}\}$. In this constellation, alt denotes the orbital altitude above the Earth's surface, inc is the angle of inclination relative to the equator, N_{orb} represents the number of orbital planes, and N_{sat} is the number of satellites per plane.

B. Dynamic Network Topology and Periodic Disconnection

This section models the network topology based on the constellation aforementioned. We represent the network topology as an undirected graph $G(V, E)$, where V denotes the satellite nodes. Each satellite V is identified by two indices: the orbital index i and the satellite index within the orbital plane j , i.e., $\{v_{i,j} \in V \mid 0 \leq i < N_{orb}, 0 \leq j < N_{sat}\}$, and the set E represents an ISL. In the mesh topology, each satellite is equipped with four laser terminals.

Due to the dynamic nature of the satellite topology, we adopt a time-varying topology model. This implies that, at each time slice, the positions of the satellites are updated. Considering the dynamicity of the satellite nodes, we also account for the impact of these changes on the edges. In other words, the movement of the satellite nodes causes changes in the positions of ISLs. Besides, the length of intra-plane ISLs [20] is given by Eq. (1), and the length of inter-plane ISLs is shown in Eq. (2). M represents the number of satellites per orbit, lat represents the latitude of satellites, RI represents the difference $RAAN$ between adjacent orbits, and R represents the radius of the Earth.

$$L_{intra} = \sqrt{2} R \sqrt{1 - \cos\left(\frac{360^\circ}{M}\right)}, \quad (1)$$

$$L_{inter} = \sqrt{2} R \sqrt{1 - \cos(RI)} \cos(lat). \quad (2)$$

As can be observed, the length of inter-plane ISLs is time-varying, and the corresponding changes in length will affect the propagation latency of the service. Therefore, the propagation latency is defined as the transmission path length divided by the speed of light, and the average propagation latency is then calculated as the mean of the propagation latency across all time slices, as shown in Eq. (3):

$$Latency_{avg} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_{i=1}^{N_t} L_{t,i} \{L_{t,i} \in \text{transmission path}\}}{c}. \quad (3)$$

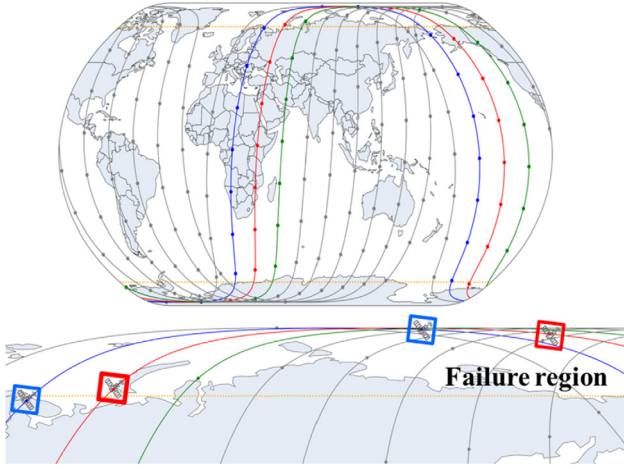


Fig. 1. Dynamic network topology and periodic disconnection.

Among them, $Latency_{avg}$ denotes the average propagation latency. T represents the total number of time slices. N_t is the number of nodes on the path in the t th time slice. $L_{t,i}$ represents the propagation distance of the i th node in the t th time slice. The condition $\{L_{t,i} \in \text{route path}\}$ checks whether the node $L_{t,i}$ belongs to the route path. Finally, c is the speed of light.

Our work focuses on predictable failures in optical satellite networks, with ISL interruptions in polar regions serving as an example in the proposed model, as shown in Fig. 1. In these regions, the limited pointing capability of the satellite laser payload can lead to ISL disconnections when the required azimuth exceeds the rotational limits. Such interruptions typically manifest as disconnections of inter-plane ISLs when a satellite's latitude exceeds 70° [21,22]. Furthermore, since the failures occur independently in each operational cycle of the satellite, the interruption events are memoryless.

C. Proactive Connectivity Handover Control Procedure

In the optical satellite network, satellites serve as transport nodes equipped with both transmission and processing capabilities, with ISLs acting as the physical transmission medium. In the centralized controller, path computation is performed, and the path information will be sent to the head node of the connection path. In the distributed control plane, path configuration information is disseminated hop-by-hop through a distributed signaling protocol (e.g., RSVP).

Based on the above control procedure, we propose a proactive connectivity handover procedure, and Fig. 2 illustrates its key differences from traditional failure recovery mechanisms. In traditional optical satellite network failure management [23] [Fig. 2(a)], upon ISL disconnections, processes such as failure detection, notification, path computation, and new path establishment are required. During the failure detection procedure, CS denotes the *channel status*, which is used to convey channel failure status information, and CSA denotes the *channel status acknowledgment*, which is used to confirm the reception of such information.

In contrast, the proactive handover mechanism predicts upcoming ISL disconnection and pre-calculates the switching time, as shown in Fig. 2(b). When the switching time is reached, the controller computes the handover path and sends the path message to the source node in the transport plane. The source satellite then forwards this message hop by hop along the new path, allowing each intermediate node to extract and record the scheduled switching time [24], so that the service is switched to the new path before the original ISL disconnects. Then, an establishment acknowledgment is returned to confirm that the new path has been successfully established. As a result, the proactive recovery mechanism effectively reduces interruption duration by eliminating the failure detection, failure notification, and path calculation latency.

D. Gravity-Based Traffic Model

To evaluate the performance of optical satellite networks, we adopted a gravity-based traffic model to simulate real-world user traffic scenarios [25]. Besides, the gravity-based traffic model accounts for the visibility between satellites and ground stations and assumes sufficient bandwidth for feeder links. Therefore, traffic from ground stations is directly mapped to satellites without considering uplink/downlink capacity constraints. Due to the large coverage area of satellites, "cities" serve as the basic unit for modeling global communication traffic [26]. The volume of traffic between cities $T_{i,j}$ is defined in Eq. (4), where P_i and P_j represent the populations of cities i and j , $D_{i,j}$ is the geographical distance between cities, and A is a gravitational constant:

$$T_{i,j} = A \cdot \frac{P_i \cdot P_j}{D_{i,j}}. \quad (4)$$

Finally, we model the arrival and departure of inter-city traffic using a Poisson process. Each connection request is characterized by the source city, destination city, and required bandwidth. Specifically, the arrival of connection requests follows a Poisson process [27]. In practical scenarios, a higher arrival rate λ generates denser traffic requests. The service duration is governed by the departure rate μ , which represents the average number of services completed per unit time.

The mapping between cities and satellites is based on the geographic coverage area of the satellite. R_{cov} is the satellite's coverage radius, alt represents the satellite's orbital altitude, and θ_{ele} is the minimum elevation angle, as shown in Eq. (5). A satellite is considered accessible to a city if the horizontal distance between them is within this radius.

$$R_{cov} = R \left(\frac{\pi}{2} - \theta_{ele} - \sin^{-1} \left(\frac{R \cos \theta_{ele}}{R + alt} \right) \right). \quad (5)$$

E. Problem Statement

In dynamic optical satellite networks, when a service interruption occurs due to ISL failure, classical reactive handover approaches migrate the service to a new path. In such a migration process, the network control plane needs to identify and localize ISL failure and further establish a new connection. Until the new connection is established, the service will be

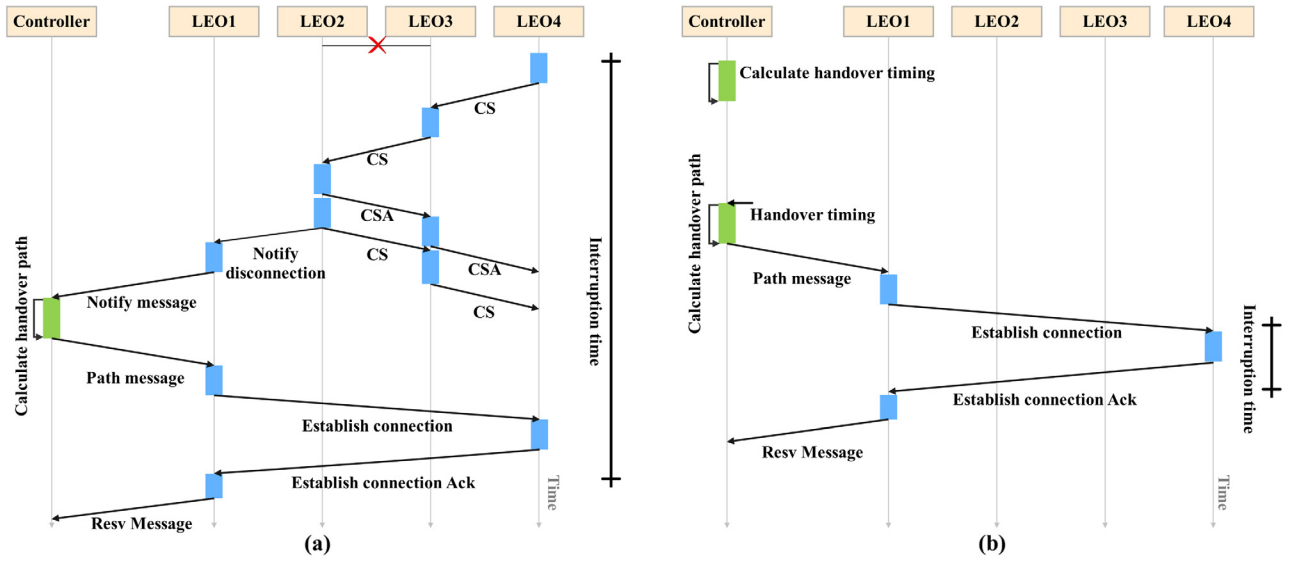


Fig. 2. Proactive connectivity handover control procedure: (a) post-failure handover control procedure and (b) proactive handover control procedure.

unavailable, and the length of the unavailable duration is a key metric for quality of service.

Thanks to the deterministic orbiting nature of satellite networks, some failures of ISLs are predictable. Accordingly, this work tries to reduce the length of the unavailable duration by addressing the connectivity handover problem in a proactive manner. However, proactive handover is more complicated than reactive handover because (1) we need to determine the connectivity switching time instead of just waiting till the connectivity is interrupted, and (2) we need to determine the new path for accommodating the service connectivity.

Therefore, we will study the proactive handover problem by addressing two sub-problems: (1) proactive connectivity switching time selection and (2) proactive connectivity switching path computation.

3. PROACTIVE CONNECTIVITY HANDOVER ALGORITHM

In this section, we first introduce the time-to-disconnection risk as a novel ISL attribute. Based on this, we design the trigger switching time and path selection, and ultimately develop a comprehensive proactive handover strategy that covers the entire end-to-end service transmission process.

A. Time-to-Disconnection Risk

This algorithm addresses scenarios involving predictable ISL failures, with its core objective being the quantitative characterization of ISL connectivity states. To this end, this work proposes the concept of time-to-disconnection risk (TDR), which serves as the foundation for a proactive handover strategy.

In the context of predictable disconnection scenarios, we analyze the case of periodic ISL disconnections in polar regions. According to the literature [8], when the absolute

value of the latitude of a satellite exceeds 70° , the corresponding ISLs are considered disconnected, as shown in Eq. (6). Meanwhile, considering the dynamic nature of the satellite network, even if the failure region is fixed, ISLs currently in a connected state may enter a disconnection state at a future time due to the continuous motion of satellite platforms. Furthermore, for different ISLs in a connected state, the risk of disconnection also varies at the same moment:

$$\text{ISLs Status} = \begin{cases} \text{disconnection} & \text{if } |\text{lat}| \geq 70^\circ \\ \text{connection} & \text{if } |\text{lat}| < 70^\circ \end{cases} \quad (6)$$

Based on spatiotemporal motion patterns and ISL disconnection modes, we propose a novel metric termed the TDR. This represents the risk of disconnecting for ISL i at time t . For disconnected ISLs, the value is set to 1. For connected ISLs, the risk is determined by the remaining time until the next disconnect. To facilitate a unified evaluation of risk levels across all ISLs in the constellation, we adopt the time interval between the next and previous disconnections. Therefore, $\text{TDR}_i(t)$ is defined as

$$\text{TDR}_i(t) = 1 - \frac{T_i^{\text{next}} - t}{T_i^{\text{next}} - T_i^{\text{previous}}}, \quad (7)$$

where t represents the current time for ISL i , T_i^{next} denotes the next disconnection time, and T_i^{previous} denotes the previous disconnection time. Based on the spatiotemporal motion pattern and the predictability of disconnections, these parameters (T_i^{next} and T_i^{previous}) can be obtained. Additionally, since intra-plane ISLs do not experience this periodic disconnection, their TDR values are 0. For inter-plane ISLs, the closer the TDR value is to 0, the lower the risk of disconnection; conversely, the closer the TDR value is to 1, the higher the risk of disconnection.

Since TDR depends solely on the predictability of ISL failures, this model is applicable to all types of predictable failures.

For instance, the sun outage failure timing can be predicted from satellite and sun positions, allowing T_i^{previous} and T_i^{next} in the TDR update formula to be readily determined [6].

B. Proactive Connectivity Switching Time Selection

The first challenge for a proactive handover algorithm is how to select the switching time: it must be designed to complete the handover before ISL disconnection occurs, while simultaneously avoiding unnecessary service interruption due to premature or excessive handovers.

Based on the aforementioned objectives, we propose the following three design principles: First is a TDR-triggered proactive handover mechanism. Since TDR directly characterizes the risk of ISL disconnection, the condition for triggering proactive handover can be set as the TDR reaching a pre-defined threshold. This design is justified for two primary reasons: on the one hand, setting the threshold to a value less than 1 ensures handover is completed before the actual ISL disconnection. On the other hand, this mechanism avoids triggering unnecessary handovers when the ISLs are still far from actual disconnection, thereby reducing the potential increase in handover frequency inherent to proactive strategies. Second, within the description in Section 2, the control procedure allows for the control plane to centrally compute the switching time, while the transport plane reliably executes the corresponding connectivity handover operations, thereby ensuring the feasibility of the proactive handover mechanism. Third, the switching time is mathematically deterministic. As established by Eq. (7), there exists a deterministic functional relationship between the selection of the switching time t and the TDR. Given a TDR, the corresponding t can therefore be directly derived via an explicit analytical expression.

Based on the aforementioned analysis, we propose a proactive switching timing selection mechanism based on TDR. First, a threshold for handover is established, denoted θ_{handover} . This threshold corresponds to a specific TDR value. Consequently, when the TDR of ISLs reaches θ_{handover} , the corresponding switching time instant t_{handover} can be derived according to Eq. (7), with its explicit expression given in Eq. (8). As illustrated in Fig. 3, once the TDR of an ISL attains the handover threshold θ_{handover} , it indicates that the system

has reached the proactive switching time t_{handover} . Notably, the accuracy of failure prediction directly affects the TDR value and thus the proactive switching timing. $\text{TDR_threshold} < 1$ provides a time margin before the actual disconnection. This buffer helps absorb moderate prediction errors: if the predicted disconnection time is slightly earlier or later than the actual failure, the handover is still completed before service interruption:

$$t_{\text{handover}} = T_i^{\text{next}} - (1 - \theta_{\text{handover}}) \cdot (T_i^{\text{next}} - T_i^{\text{previous}}). \quad (8)$$

C. Proactive Connectivity Handover Path Computation

The second challenge for a proactive handover algorithm is how to design the handover path: it must be designed to reduce the number of handovers throughout the entire transmission process.

Based on the aforementioned objectives, we propose the following two design principles: First is a handover path computation mechanism based on TDR. ISLs that are about to disconnect in a handover path can significantly increase the interruption rate. Therefore, by setting an ISL selection threshold, ISLs that are about to disconnect can be proactively avoided, thereby reducing unnecessary handovers. Second is a dynamic threshold adjustment mechanism tailored to service. Due to the asynchrony in the remaining transmission times of different services, a fixed threshold cannot meet the requirements of all services: services with longer remaining transmission times need to select ISLs with a longer time until the next disconnection, whereas services with shorter remaining transmission times can tolerate a shorter time until the next ISL disconnection. Accordingly, the threshold should be dynamically adjusted based on the remaining transmission time of each service to achieve an adaptive balance between reliability and resource efficiency.

Based on the aforementioned analysis, we propose a TDR and RTT-based dynamic handover path decision handover mechanism (TRDH). First, our previous work has introduced a path selection threshold [15], denoted as TDR-threshold, and applied it to transmission path selection during service

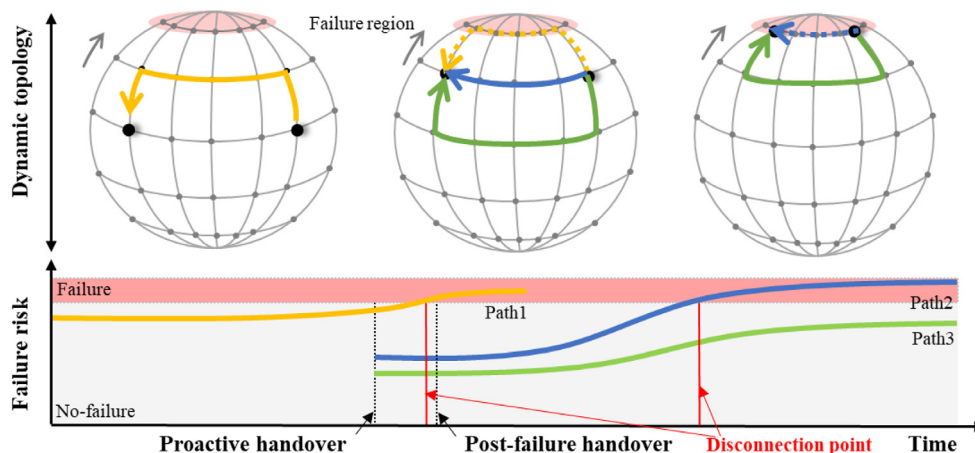


Fig. 3. Proactive connectivity switching timing selection strategy.

handover. When a connectivity handover is required, edges E whose TDR values exceed the threshold are removed from the original fully connected undirected graph $G(V, E)$, resulting in a modified graph G' , as shown in Eqs. (9), (10), and (11):

$$E_{\text{removed}} = \{e \in E \mid \text{TDR}(e) > \text{TDR_threshold}\}, \quad (9)$$

$$E' = E \setminus E_{\text{removed}} = \{e \in E \mid \text{TDR}(e) \leq \text{TDR_threshold}\}, \quad (10)$$

$$G' = (V, E'). \quad (11)$$

Furthermore, intra-plane ISLs are not removed because their TDR value is 0, thereby ensuring full connectivity within each orbital plane. Meanwhile, since satellites operate in circular orbits and the TDR of the inter-plane ISL is latitude-dependent, there always exist connectable inter-plane ISLs between adjacent orbital planes at any given time. Consequently, the graph G' guarantees that a valid connected path can be established between any source and destination satellite.

Second, to accommodate asynchronous service, we introduce a threshold interval $[r_0, r_1]$ and dynamically adjust the TDR_threshold according to Eq. (12), where the remaining transmission time ratio (rtt_ratio) denotes the ratio of the remaining transmission time to the total transmission time:

$$\text{TDR_threshold} = r_0 + \text{rtt_ratio} \times (r_1 - r_0). \quad (12)$$

In summary, when performing handover path computation for different services, each service first calculates its TDR_threshold according to Eq. (12). Based on this, the updated network topology graph G' is derived from Eq. (11). Subsequently, the shortest path algorithm is executed on G' to determine the optimal handover path, as illustrated in Fig. 4.

Finally, we illustrate the impact of TDR through an example. As shown in Fig. 4, both algorithms achieve handover to an available path. However, due to the dynamic topology, the path selected by the conventional shortest path (SP) algorithm based on Dijkstra's shortest-path principle re-enters the failure region shortly after handover, causing a secondary interruption. In contrast, the TRDH algorithm excludes high-risk ISLs during path selection, thereby preventing further interruptions before transmission completion.

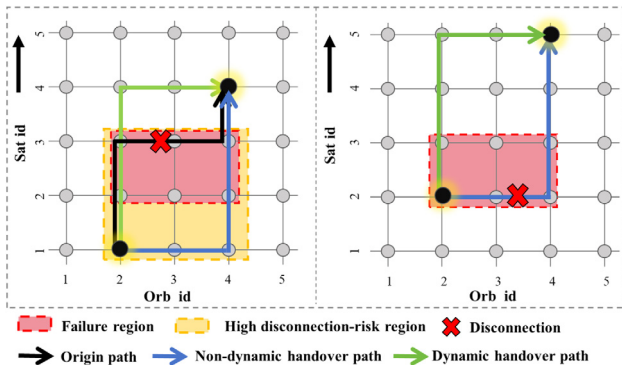


Fig. 4. TRDH algorithm.

Algorithm 1. Proactive Connectivity Handover Algorithm

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1: Input: Graph  $G = (V, E)$ , threshold interval  $[r_0, r_1]$ , time slices  $\mathcal{T}$ , set of services  $\mathcal{S} = \{(\text{src}_s, \text{dst}_s)\}_{s=1}^{|\mathcal{S}|}$ 
2: Output: Switching time and handover paths for affected services
3: for each time slice  $t \in \mathcal{T}$  do
4:   Compute  $\text{TDR}_i(t)$  for all  $E$  using Eq. (7)
5:   for each service  $s \in \mathcal{S}$  do
6:     Calculate switching time  $t_{\text{handover}}(s)$  using Eq. (8)
7:     if  $t_{\text{handover}}(s) \geq t$  then
8:       Calculate  $\text{rtt\_ratio}$  for service  $s$ 
9:       Calculate  $\text{TDR\_threshold}_s$  using Eq. (12) based on  $[r_0, r_1]$  for service  $s$ 
10:      Initialize  $G'(s) = G$ 
11:      for each ISL  $i \in E$  do
12:        if  $\text{TDR}_i(t) > \text{TDR\_threshold}_s$  then
13:          Remove  $i$  from  $G'(s)$  using Eqs. (9), (10), (11)
14:      Apply Dijkstra's Algorithm to  $G'(s)$ :
15:       $\text{handover\_path}_s = \text{Dijkstra}(G'(s), \text{src}_s, \text{dst}_s)$ 

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D. Proactive Connectivity Handover Algorithm

Based on the switching timing selection and handover path computation strategies, we propose a proactive handover algorithm, namely, the proactive TRDH (PTRDH) algorithm, whose complete procedure is described in Algorithm 1.

For each time slice, it first computes the TDR for all ISLs using Eq. (7). For each service, it calculates the switching time using Eq. (8). If the current time slice t has reached the proactive switching time, the control plane proceeds to compute the rtt_ratio for the affected service and determines TDR_threshold_s using Eq. (12) based on the threshold interval $[r_0, r_1]$. It then removes ISLs whose TDR exceeds TDR_threshold_s in $G'(s)$. Dijkstra's algorithm is then executed on $G'(s)$ to obtain the handover path for that service.

E. Complexity Analysis

The computational complexity of the PTRDH algorithm mainly comes from TDR calculation and path computation using Dijkstra's algorithm. The complexity of $\text{TDR}_i(t)$ calculation is $O(|E_{\text{inter}}|)$ and the complexity of Dijkstra's algorithm is $O((|V| + |E_{\text{inter}}|) \lg(|V|))$, where $|V|$ is the number of nodes in the slice topology $G_s = (V, E_{\text{inter}})$, and $|E_{\text{inter}}|$ is the number of inter-plane ISLs. In each slice, there are N services requiring handover, where N is a constant, and $|E_{\text{inter}}| = O(|V|)$. Consequently, the final computational complexity of the PTRDH algorithm can be expressed as $O(|V| \lg(|V|))$.

In the highly dynamic satellite networks, frequent ISL interruptions might cause parallel service handovers. Fortunately, in our proactive approach, path computation is conducted in advance, which does not require immediate determination of paths for post-failure restoration. This allows batch processing of handover tasks, alleviating the real-time pressure on the controller. Moreover, recent advances in high-performance computing, such as using CUDA, can achieve $10\times$ faster path computing speed than the traditional solutions [28], which significantly improve scalability.

4. SIMULATION RESULTS AND ANALYSIS

A. Simulation Settings

In the simulation, we adopt the orbital parameters of OneWeb, specifically the Walker-Delta orbital configuration with an inclination angle of 87.9° , and constellation parameters of $\{780 \text{ km}, 87.9^\circ, 40, 30\}$. Consequently, the total number of satellites is 1200, with each satellite equipped with four laser terminals to establish ISLs, forming two intra-plane ISLs and two inter-plane ISLs in a full grid-mesh topology.

The connection between the satellite and the ground station is established, with the minimum elevation angle θ_{ele} set to 25° . The simulation assumes that each satellite can accommodate a sufficient number of uplink and downlink connections. While uplink/downlink bandwidth constraints may affect overall network capacity (e.g., in terms of blocking probability or throughput), they primarily influence the number of services that can be provisioned rather than the performance of successfully established connections. Since the proposed proactive handover algorithm aims to improve the performance of successfully provisioned services (e.g., maximum reliable transmission time and end-to-end latency), this simplification enables a focused analysis of the algorithm's effectiveness in reducing interruption time without compromising the validity of the comparative results.

The parameters associated with network traffic, including bw , μ , and λ , are shown in Table 1. It is important to note that the proposed gravity-based traffic model is employed to generate workload weights between different pairs of cities. Consequently, each request's source-destination city pair is randomly selected based on the corresponding weight of all candidate city pairs. The source-destination city pair is then mapped to a satellite pair using the random city-to-satellite mapping method described in Section 2.D. The bandwidth requirement for each request is generated randomly within the range of $[1, 50]$ Mbps, following a uniform distribution.

Moreover, the TDR_threshold is tunable and configurable, which can be dynamically adjusted based on network requirements. For example, reliability-sensitive service can adopt a smaller threshold than a normal service to reserve a longer risk buffer time. Additionally, the threshold value is set following two principles: first, it is set to a value less than 1 to proactively avoid service interruption; second, it accounts for service asynchrony, where services with longer remaining transmission

Table 1. Parameter Notations and Their Values

Notation	Value	Note
N_{orb}	40	Total number of orbits
N_{sat}	30	Number of satellites per plane
alt	780 km	The altitude of the satellite
inc	87.9°	Inclination
θ_{ele}	25°	Minimum elevation angle
λ	$[30, 80]$	Arrival rate
μ	0.01	Departure rate
bw	$[1, 50]$ Mbps	Bandwidth demand
TDR_threshold	$[0.7, 0.9]$	ISL risk threshold
SDSF	$[0.625, 1.25]$	Service duration scale factor
θ_{handover}	0.9	Threshold for switching time

Table 2. Handover Algorithm

Notation	Note
SP	Dijkstra's algorithm for shortest path
TBH	TDR-based handover algorithm
$TRDH$	TDR-based risk-threshold dynamic handover
PSP	Proactive handover-based SP
$PTBH$	Proactive handover-based TBH
$PTRDH$	Proactive handover-based TRDH

times require more conservative thresholds. So, the TBH algorithm is tested with three different TDR values: 0.7, 0.8, and 0.9. The TDR_threshold in the TRDH algorithm is calculated within the range of 0.7 to 0.9.

In the optical satellite network with the aforementioned settings, this work simulates several connectivity handover algorithms, as shown in Table 2. To demonstrate the impact of TDR-based handover path selection, we simulated the SP algorithm and the TBH algorithm. Furthermore, this work investigates the influence of the TDR selection on network performance by comparing the fixed TDR-based TBH algorithm with the dynamically adjusted TDR-based TRDH algorithm. In addition, to further highlight the necessity of proactive handover algorithms, we simulated the PSP, PTBH, and PTRDH algorithms to explore the advantages of proactive handover.

To evaluate the impact of the proposed algorithms on network performance, we design two different simulation scenarios and conduct tests under various horizontal axis settings, as detailed below.

- **Arrival rate:** We generate random connection requests following a Poisson process, where the arrival rate λ is in the range of $[30, 80]$, and the mean service rate μ is fixed at 100.
- **Service duration scale factor (SDSF):** The SDSF is the service duration scale factor, which represents the duration scale of services in the network. In the simulation, the SDSF is used to adjust the average service duration, with a range of $[0.625, 1.25]$ and a step size of 0.125, in order to analyze the performance of services under different service duration scales.

Frequent interruptions can negatively impact the reliability of satellite networks [13]. Therefore, we consider the interruption rate as a key performance metric. The interruption rate is defined as the ratio of the total number of handovers across all services to the total number of slices across all services, where the continuous time is divided into discrete time slices, and each time a service performs a handover, it is counted as one handover slice. A lower interruption rate corresponds to higher network reliability. Moreover, under the same interruption rate, a longer maximum reliable transmission time can significantly enhance the user experience [29]. The maximum reliable transmission time refers to the maximum time interval between two successive handovers, indicating the longest duration for maintaining stable transmission. As a result, we use the maximum reliable transmission time as another important performance metric for evaluation. This work focuses on improving the reliability of satellite networks as an objective, while systematically evaluating its impact on other network

performance metrics. Thus, we compare the performance of different algorithms in terms of propagation latency.

In conclusion, this work will conduct a comprehensive performance evaluation of optical satellite networks. Various performance metrics will be assessed, including the average interruption rate, maximum reliable transmission time, and propagation latency. These evaluations will be performed under different arrival rates and service durations.

B. Result for TRDH

In this section, simulations of the SP, TBH, and TRDH algorithms are conducted, and a comparative analysis is presented regarding the performance of these algorithms in terms of interruption rate, maximum reliable transmission time, and propagation latency under varying arrival rates and SDSF conditions. We compare and evaluate the performance of the SP, TBH, and TRDH algorithms.

Figure 5 illustrates the variation of the interruption rate, maximum reliable transmission time, and propagation latency with respect to the arrival rate. In the simulation, the arrival rate is set within the range of [30, 80] with a step size of 10. The dashed lines represent the average values of the corresponding metrics, and the numerical labels on the right indicate these averages.

Figure 5(a) illustrates the results of the interruption rate for the SP, TBH, and TRDH algorithms. The SP algorithm exhibits the highest interruption rate, with an average value of 82.4%, while the TRDH algorithm achieves the lowest average interruption rate at approximately 48.6%. The average interruption rate is ranked from smallest to largest as follows: TRDH < TBH-0.7 < TBH-0.8 < TBH-0.9 < SP. In addition, compared to the SP algorithm, TRDH can reduce the interruption rate by 41.02%. Subsequently, the TBH algorithm also plays a role in reducing the interruption rate: when the TDR is set to 0.7, it achieves the best performance, reducing the average interruption rate by 40.78%. When the TDR is set to 0.8, the average interruption rate is reduced by 38.71%, and when the TDR is set to 0.9, it performs the worst but still reduces the average interruption rate by approximately 30.10%.

In addition, several other conclusions can be drawn: (1) It is necessary to consider the TDR of ISLs, as this prevents services from handover to high-risk ISLs that are about to disconnect,

thereby reducing the interruption rate. (2) Within the range of TDR_threshold values from 0.7 to 0.9, the smaller the TDR_threshold, the better the interruption rate performance. This is because the TDR_threshold directly affects the tolerance to disconnection risks for ISLs. The smaller the TDR, the lower the tolerance to ISL disconnection. This makes the handover path less likely to fail and reduces the interruption rate. (3) Notably, the TRDH algorithm, which considers both TDR and rtt_ratio, performs exceptionally well, achieving results comparable to TBH-0.7. This indicates that, by considering TDR alongside the rtt_ratio of the service, the performance of handover path selection is not compromised. Taken together, handover algorithms that consider ISL disconnection risk significantly reduce the interruption rate, and the selection and adjustment of TDR_threshold play a crucial role in optimizing network performance.

Figure 5(b) illustrates the results of the maximum reliable transmission time with respect to the arrival rate. In Fig. 5(b), the SP algorithm performs the shortest maximum reliable transmission time, with an average value of 850.45s, while the TRDH algorithm performs the best, with an average of 903.01s. For the TBH algorithm, different TDR settings lead to varying results. When TDR = 0.7, the maximum reliable transmission time is 901.84s; when TDR = 0.8, the maximum reliable transmission time is 901.33s; and when TDR = 0.9, the average maximum reliable transmission time is 893.33s. The ranking of maximum reliable transmission time from shortest to longest is as follows: SP < TBH-0.9 < TBH-0.8 < TBH-0.7 < TRDH.

In addition, several other conclusions can be drawn: (1) Within the TDR_threshold range of 0.7 to 0.9, the lowest TDR_threshold (the minimum tolerance for disconnection) results in the highest maximum reliable transmission time. (2) Furthermore, the TRDH algorithm, with its dynamic TDR_threshold strategy that accounts for the service's rtt_ratio, performs more effectively, demonstrating that dynamic TDR_threshold selection is better suited to the asynchronous nature of service durations in the optical satellite network. Therefore, handover algorithms considering ISL disconnection risk can improve the maximum reliable transmission time, and selecting an appropriate TDR_threshold is critical. A fixed TDR_threshold may fail to effectively address asynchrony in service, while algorithms that dynamically adjust the TDR_threshold are more appropriate.

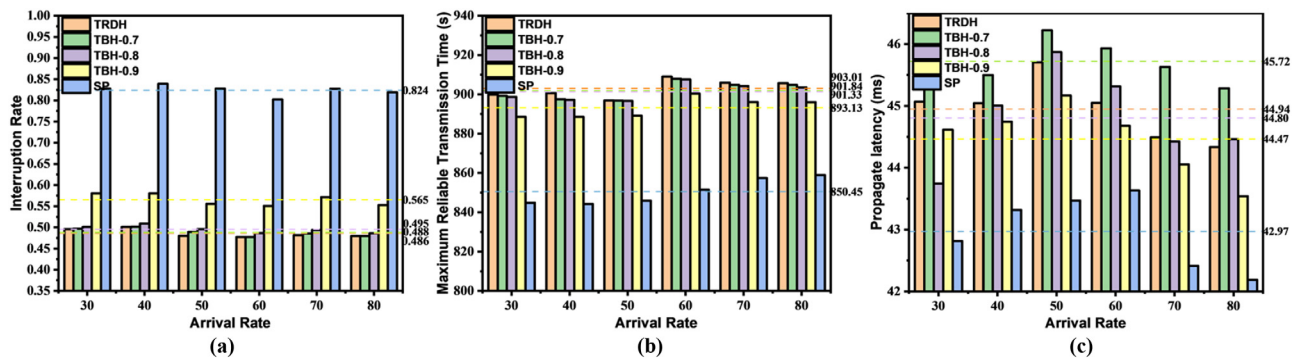


Fig. 5. Results of the TRDH algorithm under different arrival rates: (a) interruption rate, (b) maximum reliable transmission time, and (c) propagation latency.

Figure 5(c) illustrates the results of propagation latency with respect to the arrival rate. The propagation latency increases in the following order: SP < TBH-0.9 < TBH-0.8 < TRDH < TBH-0.7. Analysis indicates that the increase in propagation latency for the TBH and TRDH algorithms is primarily due to the exclusion of high-risk ISLs during path selection, resulting in longer lengths. Furthermore, since the length of inter-plane ISLs varies over time, the propagation latency in this work is calculated based on time-slot updates based on Eq. (3). As a result, algorithms with higher interruption rates update the shortest path more frequently, while algorithms with lower interruption rates tend to maintain the original path even as its length increases. This is the main reason for the increase in propagation latency.

Notably, the TRDH algorithm performs the best in terms of both average interruption rate and maximum reliable transmission time, while reducing propagation latency by 1.7% compared to the second-best-performing TBH-0.7 algorithm. Therefore, the dynamic TDR selection strategy of the TRDH algorithm proves to be more advantageous, as it not only enhances the stability of the optical satellite network but also reduces the trade-off cost on propagation latency, resulting in the most superior overall performance.

Figure 6 illustrates the results of the interruption rate, maximum reliable transmission time, and propagation latency with respect to the SDSF. In the simulation, the range of the SDSF is set to [0.625, 1.250] with a step size of 0.125. Furthermore, to ensure the accuracy of the results, the arrival rate is set to 30 for all different SDSFs.

Figure 6(a) illustrates the results of the average interruption rate with respect to the SDSF. The SP algorithm exhibits the highest interruption rate, with an average value of 77.6%. The TRDH algorithm continues to perform the best, achieving the lowest average interruption rate of 45.5%, resulting in a reduction of 41.37% compared to the SP algorithm. The ranking of interruption rates from lowest to highest is as follows: TRDH < TBH-0.7 < TBH-0.8 < TBH-0.9 < SP. Here, some conclusions can be drawn: considering the TDR significantly reduces the interruption rate; the TRDH algorithm outperforms the others, showing that incorporating *rtt_ratio* enhances TDR performance and optimizes the interruption rate.

In addition, several other conclusions can be drawn: (1) As the SDSF increases, the interruption rate also increases. This is

due to the periodic nature of ISL failures, which causes the ISL disconnection rate to remain constant. As a result, services with longer durations experience more ISL disconnections, leading to a higher interruption rate. Therefore, services with longer durations should prioritize handover algorithms that enhance stability. (2) As the service duration increases, the advantages of the TRDH algorithm become more pronounced. For example, when the SDSF is 0.625, the TRDH algorithm reduces the interruption rate by 39.06% compared to the SP algorithm; when the SDSF is 1.25, the reduction increases to 41.86%. Therefore, for services with longer durations, designing appropriate handover algorithms is crucial, and their effectiveness becomes more pronounced.

Figure 6(b) illustrates the results of maximum reliable transmission time with respect to the SDSF. The TRDH algorithm outperforms others, achieving the longest maximum reliable transmission time, with an average of up to 910.15 s. Specifically, the ranking of the maximum reliable transmission time is as follows: TRDH > TBH-0.7 > TBH-0.8 > TBH-0.9 > SP. Here, some conclusions can be drawn: handover algorithms that incorporate TDR significantly enhance the maximum reliable transmission time; and the TRDH algorithm optimizes the maximum reliable transmission time, indicating that *rtt_ratio* incorporation does not negatively affect the algorithm's performance.

In addition, several other conclusions can be drawn: (1) As the SDSF increases, the maximum reliable transmission time also increases, primarily because the SDSF leads to longer service durations. (2) However, the growth rate of the maximum reliable transmission time is slower than that of the SDSF, as longer transmission durations are associated with more frequent service interruptions. (3) Specifically, when the SDSF is 1.25, the performance improvement reaches 97.12 s; when the SDSF is 0.625, the improvement is 48.04 s. These results indicate that the TRDH algorithm is more effective for services with longer durations. Therefore, the TRDH algorithm demonstrates stronger performance optimization capabilities for services with longer transmission times.

Figure 6(c) illustrates the results of propagation latency with respect to the SDSF. The TBH-0.7 algorithm exhibits the highest propagation latency, with an average value of 43.38 ms, while the SP algorithm demonstrates the lowest propagation latency, with an average value of 39.59 ms. The propagation latency ranked from lowest to highest is as follows:

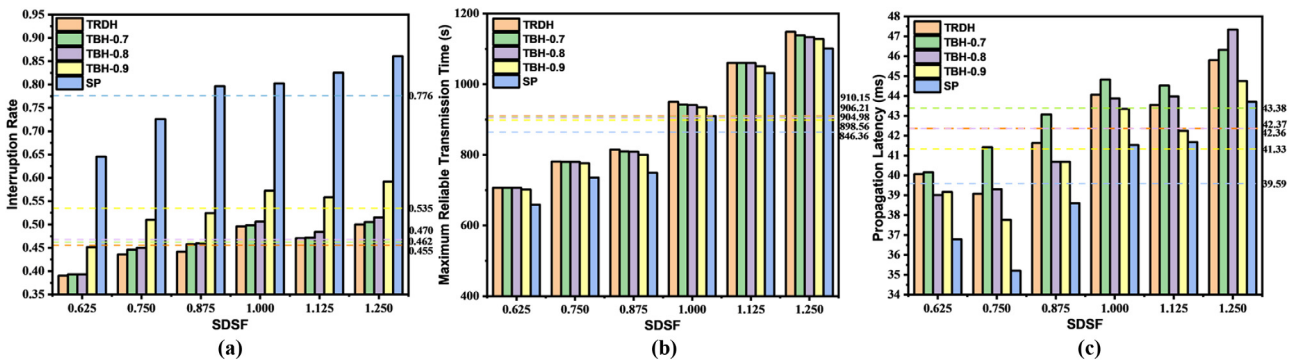


Fig. 6. Results of the TRDH algorithm under different SDSFs: (a) interruption rate, (b) maximum reliable transmission time, and (c) propagation latency.

$SP < TBH-0.9 < TBH-0.8 < TRDH < TBH-0.7$. Compared with the TBH-0.7 algorithm, the TRDH algorithm not only further reduces the interruption rate, but also minimizes the increase in latency. Its average propagation latency is 42.37 ms, representing a reduction of 2.38% in propagation latency. This advantage comes from the consideration of both the TDR and the rtt_ratio of services by the TRDH algorithm.

In addition, several conclusions can be drawn: (1) As the SDSF increases, the propagation latency also slightly increases. This is due to the time-varying nature of inter-plane ISL lengths: ISLs moving from high latitudes to low latitudes become longer, while those moving from low latitudes to high latitudes become shorter. (2) In addition, high-TDR ISLs typically move from low to high latitudes and are excluded from the handover path selection. Consequently, the remaining ISLs in the selected path result in increased propagation latency over time. Therefore, as the SDSF increases, the propagation latency also increases.

C. Result for PTRDH

In this section, we introduce the proactive handover strategy based on the SP, TBH, and TRDH algorithms, resulting in the formation of the PSP, PTBH-0.7, PTBH-0.8, PTBH-0.9, and PTRDH algorithms. We analyze the performance of the above algorithms in terms of interruption rate, maximum reliable transmission time, and propagation latency.

Figures 7 and 8 show the results of the interruption rate under different arrival rates and SDSF values. In Fig. 7, the performance ranking of the handover algorithms is consistent with the results without the proactive handover strategy. Furthermore, the interruption rate of the PTRDH algorithm is comparable to that of the best-performing TRDH algorithm, with reductions of 41.93% and 41.94%, respectively, compared to the PSP algorithm. In Fig. 8, the average interruption rate of the PTRDH algorithm is also similar to that of the TRDH algorithm, with reductions of 40.12% and 39.96% compared to the PSP algorithm.

Based on these results, several conclusions can be drawn: the interruption rate is consistent with the performance without the proactive interruption strategy, as the path selection strategies remain the same. Notably, despite the difference in switching time between PTRDH and TRDH, both algorithms exhibit a similar interruption rate. This indicates that the PTRDH algorithm does not lose control over the ISL state due to pre-computed paths. Specifically, PTRDH avoids prematurely selecting the switching time, preventing a loss of control over future ISL states, while also enabling proactive handover in advance, further enhancing network performance.

Figures 9 and 10 show the results of the maximum reliable transmission time under different arrival rates and SDSF values. In Fig. 9, the average maximum reliable transmission time of the PTRDH algorithm is 1121.72 s, showing the best performance. Compared to the baseline SP algorithm in Fig. 5, this represents an improvement of 271.27 s, or a 31.89% increase. However, the TRDH algorithm has the worst performance with an average maximum reliable transmission time of 902.83 s. In Fig. 10, as the SDSF increases,

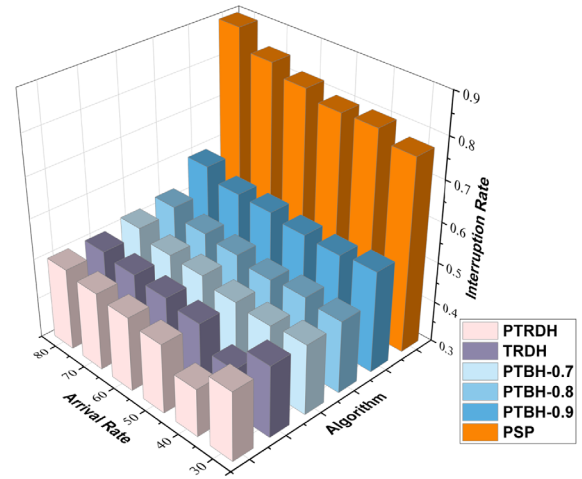


Fig. 7. Interruption rate of proactive handover algorithms versus the arrival rate.

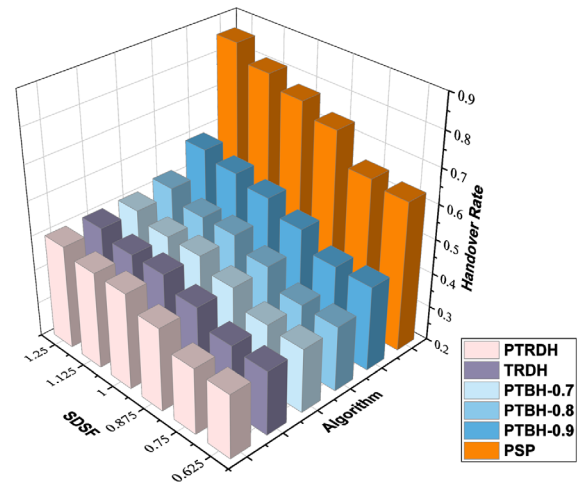


Fig. 8. Interruption rate of proactive handover algorithms versus the SDSF.

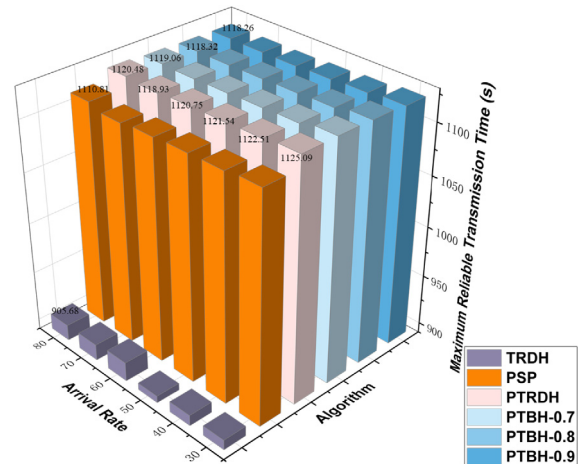


Fig. 9. Maximum reliable transmission time of proactive handover algorithms versus the arrival rate.

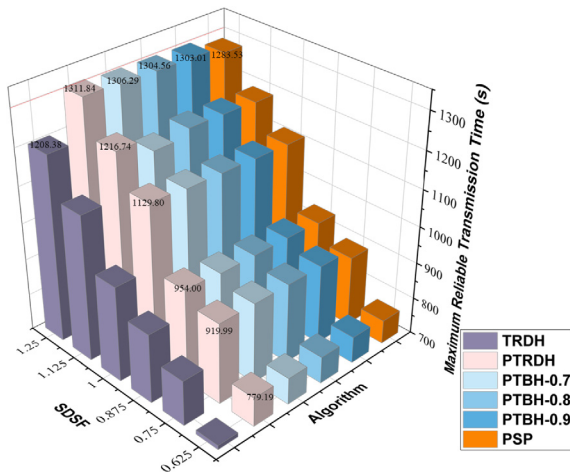


Fig. 10. Maximum reliable transmission time of proactive handover algorithms versus the SDSF.

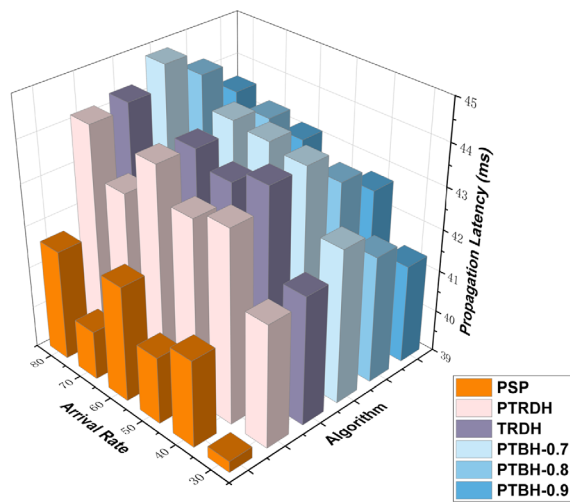


Fig. 11. Propagation latency versus the arrival rate.

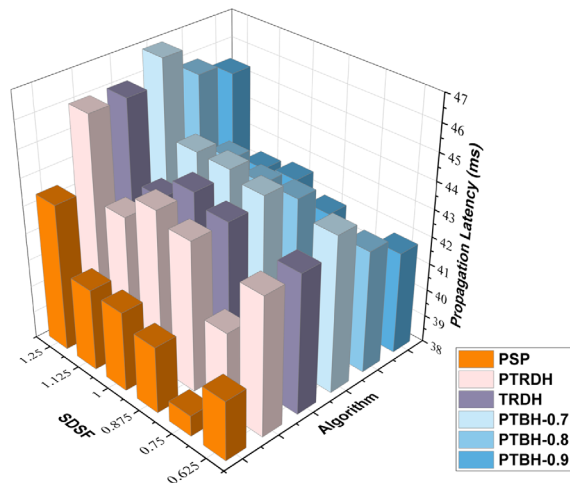


Fig. 12. Propagation latency versus the SDSF.

the PTRDH algorithm still exhibits the best performance with the average maximum reliable transmission time of 1051.92 s. Compared to the baseline SP algorithm in Fig. 6, this represents an improvement of 207.23 s, or a 24.53% increase. While the TRDH algorithm has the worst performance with an average maximum reliable transmission time of 948.46 s. Among the proactive handover algorithms, the performance ranking is as follows: PTRDH > PTBH-0.7 > PTBH-0.8 > PTBH-0.9 > PSP > TRDH.

Based on these results, several conclusions can be drawn: A comparison between the TRDH and PTRDH algorithms demonstrates that the proactive handover strategy significantly enhances performance. Specifically, it increases the average maximum reliable transmission time by approximately 218.88 s in Fig. 9 and 103.46 s in Fig. 10. The only difference between PTRDH and TRDH lies in the adoption of the proactive handover strategy. The proactive handover strategy reduces the interruption time caused by failure detection, failure identification, and failure notification, thereby maintaining the same interruption rate while reducing the occurrence of interruptions, which further enhances the maximum reliable transmission time. Therefore, the necessity of the proactive handover strategy is confirmed.

In addition, despite the adoption of the same proactive handover strategy, different path selection strategies still lead to significant performance variations. Among the aforementioned proactive handover algorithms, the PSP algorithm performs the worst, while PTRDH outperforms PSP by approximately 10.25 s in Fig. 9 and 21.39 s in Fig. 10. This is because, although both algorithms perform handovers before the failure occurs, the path selected by the PSP algorithm fails to avoid the impending disconnection ISLs, which leads to another interruption shortly after the handover. Therefore, the strategy of the handover path remains crucial to achieving optimal performance.

Figures 11 and 12 show the results of the propagation latency under different arrival rates and SDSF values. In Fig. 11, the latency ranking is PTBH-0.7 > PTRDH > TRDH > PTBH-0.8 > PTBH-0.9 > PSP, with PTRDH having an average latency of 43.68 ms and TRDH of 43.35 ms. (Note that the results shown in Figs. 6(c) and 11 are obtained from two independent simulation runs. Therefore, slight numerical differences may exist between them.) In Fig. 12, the ranking is PTBH-0.7 > TRDH = PTRDH > PTBH-0.8 > PTBH-0.9 > PSP, with both PTRDH and TRDH having an average latency of 43.42 ms.

Based on these results, several conclusions can be drawn: PTRDH and TRDH perform similarly in terms of propagation latency, which is influenced by the handover path. Both strategies employ the same path computation method, differing only in the timing of handover. Additionally, in Fig. 11, the average latency of PTBH-0.7 is 44.18 ms, which is approximately 0.83 ms higher than that of PTRDH (43.35 ms). In Fig. 12, the average latency of PTBH-0.7 is 43.96 ms, about 0.54 ms higher than PTRDH (43.42 ms). These results indicate that, although both algorithms adopt proactive handover strategies, the dynamic TDR selection scheme (PTRDH) outperforms the fixed-TDR TBH algorithm. In conclusion, to enhance the reliability of optical satellite networks while

minimizing propagation latency, the PTRDH algorithm is the optimal choice.

5. CONCLUSION

In satellite networks, predictable ISL failures cause frequent service interruptions, severely impacting network reliability. This work defined ISL risks and proposed a proactive connectivity handover algorithm (PTRDH) to reduce total interruption duration. This algorithm determine the switching time and path for each handover operation, instead of waiting till the connectivity is interrupted. We conducted simulation-based comparative analyses of several strategies: SP, TBH (fixed threshold), TRDH (dynamic threshold), and PTRDH (proactive handover). The key findings are as follows: (1) Service handovers occur very frequently in satellite networks, with a interruption rate of up to 82.4% when using the SP algorithm. (2) Introducing a TDR_threshold mechanism in path selection significantly reduces the interruption rate, achieving a maximum reduction of 41.02%. (3) The proactive handover mechanism substantially decreases interruption latency by proactive handovers, with the maximum reliable transmission time improved by up to 31.89%. (4) The dynamic risk threshold mechanism ensures a lower interruption rate while reducing trade-off cost on propagation latency by approximately 1.7%. In summary, the PTRDH algorithm can reduce the total interruption time of the optical satellite network, which plays a key supporting role in enhancing the overall reliability of optical satellite networks. In future work, we plan to introduce satellite-ground visibility constraints and extend the proactive handover strategy to space-ground integrated routing.

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